

EPPS 6302 Methods of Data Collection and Production:

Assignment 4

Webscraping 1

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What you will produce

A cleaned dataframe scraped from a static page with `rvest`, the same quantity fetched from an API for comparison, and a one-page data plan for acquiring web data in your own research.

0. Ask permission first

Before the first request, check that the path you intend to fetch is permitted, and say so in your write-up.

```
robotstxt::paths_allowed("https://en.wikipedia.org/wiki/List_of_countries_by_foreign-  
↪ exchange_reserves")
```

1. Warm up on a page built to be scraped

- Run the first block of `rvest_wiki01.R` against <https://books.toscrape.com> — a sandbox published for exactly this purpose: 1,000 books, 50 pages, no JavaScript, nobody's server to bother.
- Extract title, price and rating. Then page through five pages and bind them into one dataframe.

Learn the verbs somewhere harmless. `read_html()` → `html_elements()` → `html_text2()` / `html_attr()` / `html_table()`. Get them wrong on a sandbox, not on a live site with a rate limiter.

2. Scrape the Wikipedia table

- Use `rvest_wiki01.R` to scrape the **foreign-exchange reserves** table from Wikipedia.
- Modify the program to scrape a different Wikipedia table of your choosing. Note what broke when you changed target, and why.
- Clean the dataframe:
 - the `date` variable
 - remove unneeded rows and columns — footnote markers `[1]`, thin spaces, the multi-row header, any *Total* or *World* row

3. Now do it the right way

- The same quantity is published by the **World Bank** as indicator `FI.RES.TOTL.CD`, *Total reserves (includes gold, current US\$)*. Fetch it with the `WDI` package — one call.
- Compare the two results. Which countries differ, and why? Which vintage is each source reporting?
- In two sentences: **when is scraping the right tool?**

Wikipedia is a poor scraping target, and that is the lesson. It is volunteer-edited, has no schema, no versioning and no stated uncertainty, and the table shape changes without notice — so your parser breaks silently. It also has a documented API and bulk dumps, which means scraping it is a *choice you did not have to make*. Scrape when there is no other way in.

4. Data plan

Write **one page**: a plan to acquire web data for your own research.

- The **source**, and why that source rather than another
- What `robots.txt` and the Terms of Service permit
- Whether an **API** exists — and your justification if you scrape anyway
- Volume, rate limiting, and how you will cache
- What breaks when the page changes, and how you will notice

5. Submit

- Publish as `assign04.qmd` under **Assignments** on your website, and send the URL to the teaching assistant.